



10961 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	2	Spec RV1-col1	NIRSpec IFU Spectroscopy	(1) GALCEN
	3	Spec RV1-col2	NIRSpec IFU Spectroscopy	(1) GALCEN
	5	Spec RV1-col3	NIRSpec IFU Spectroscopy	(1) GALCEN
	4	Images2	NIRCam Imaging	(1) GALCEN
	6	Spec RV2-col1	NIRSpec IFU Spectroscopy	(1) GALCEN
	8	Spec RV2-col2	NIRSpec IFU Spectroscopy	(1) GALCEN
	9	Spec RV2-col3	NIRSpec IFU Spectroscopy	(1) GALCEN

ABSTRACT

The Galactic Center (GC) is a rich laboratory routinely delivering breakthrough discoveries to help us understand the nuclei of all galaxies. However, our picture is far from complete. Questions abound, including: (1) How often do BH-BH binaries form, merge, and grow around the SMBH, as observed by gravitational wave experiments? (2) Would black holes (BHs) swarm around the SMBH and how does their total mass compare with that of a predicted dark-matter cusp? (3) How often do stellar binaries close to the SMBH lead to hypervelocity stars, tidal disruption events, and mergers that produce puffy stellar objects? To address these questions, we require a thorough census of the binary systems within the nuclear star cluster. JWST's unique combination of depth and field of view makes it ideal for probing the binary population at the GC through a combination of astrometric wobbles, periodic photometric variations, and radial velocities. We propose Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS) to obtain multi-epoch spectroscopy and radial velocities of the central 9" (0.36 pc) and multi-epoch photometry and astrometry of the central 2' (4.8 pc). BIGCATS combined with archival JWST data will provide the most complete census of binary and compact object populations around an SMBH to answer the questions above. BIGCATS will also provide a treasury for future studies of black hole accretion, gas and stellar orbits, stellar variability, star formation history, metallicity distributions, and much more. A thorough understanding of our own GC where we can resolve stars will provide key insights applicable to all galaxies and their nuclear star clusters.

OBSERVING DESCRIPTION

We propose to obtain NIRSpec IFU spectroscopy of the central 9" (0.36 pc) and NIRCam imaging of the central 2' (4.8 pc) of our Galaxy in order to study the nuclear star cluster, its binary population, and its co-evolution with the supermassive black hole.

The NIRSpec IFU spectroscopy uses the higher-resolution G235H grating in the F170LP filter to obtain spectra from 1.6 - 3.3 microns for a 3x3

mosaic at the Galactic Center and will be repeated 2 times in Cycle 5. The spectra will be used to measure the radial velocities and metallicities of young and old stars. These observations will be used to find and confirm binary candidates through radial velocities and astrometry, measure binary orbits and orbits around the supermassive black hole, and search for black holes in binary systems around the SMBH.

The NIRCам imaging includes 2 filters (F212N, F323N) conducted once in cycle 5 to measure proper motions for stars down to $K=20$ at better than 0.25 mas/yr (20 km/s). The astrometry will be used to measure the positions, proper motions, accelerations, and orbits of stars around the supermassive black hole and to measure the astrometric wobble of stars in binary systems.

The above spectroscopy and imaging observations will be repeated in Cycle's 6 and 7 for extended time baseline.

Proposal 10961 - Targets - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	GALCEN	RA: 17 45 40.0410 (266.4168375d)	Proper Motion RA: -2.7 mas/yr	
			Dec: -29 00 28.12 (-29.00781d)	Proper Motion Dec: -5.6 mas/yr	
			Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Stellar associations, Young star clusters] Extended=YES				

Proposal 10961 - Observation 2 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Observation	Proposal 10961, Observation 2: Spec RV1-col1											Wed May 20 18:00:46 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	GALCEN	RA: 17 45 40.0410 (266.4168375d)				Proper Motion RA: -2.7 mas/yr					
	Dec: -29 00 28.12 (-29.00781d)				Proper Motion Dec: -5.6 mas/yr							
	Equinox: J2000				Epoch of Position: 2015.5							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Stellar associations, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	3	1	3.0		3.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		9			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	25	1	false	true	NONE	9	9	2512.404	
	2	G235H/F170LP	NRSRAPID	10	1	true	true	NONE	9	9	1062.94	

Proposal 10961 - Observation 2 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Special Requirements	Offset -2.91 arcsec, 0.0 arcsec 3 After 2 by 0 Days to 14 Days
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Proposal 10961 - Observation 3 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Observation	Proposal 10961, Observation 3: Spec RV1-col2											Wed May 20 18:00:46 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	GALCEN	RA: 17 45 40.0410 (266.4168375d)				Proper Motion RA: -2.7 mas/yr					
			Dec: -29 00 28.12 (-29.00781d)				Proper Motion Dec: -5.6 mas/yr					
			Equinox: J2000				Epoch of Position: 2015.5					
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Stellar associations, Young star clusters] Extended=YES										
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	3	1	3.0		3.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	25	1	false	true	NONE	9	9	2512.404	
	2	G235H/F170LP	NRSRAPID	10	1	true	true	NONE	9	9	1062.94	

Proposal 10961 - Observation 3 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Special Requirements	3 After 2 by 0 Days to 14 Days 5 After 3 by 0 Days to 14 Days
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Proposal 10961 - Observation 5 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Observation	Proposal 10961, Observation 5: Spec RV1-col3											Wed May 20 18:00:46 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpect IFU Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	GALCEN	RA: 17 45 40.0410 (266.4168375d)				Proper Motion RA: -2.7 mas/yr					
	Dec: -29 00 28.12 (-29.00781d)				Proper Motion Dec: -5.6 mas/yr							
	Equinox: J2000				Epoch of Position: 2015.5							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
Category=Stellar Cluster												
Description=[Stellar associations, Young star clusters]												
Extended=YES												
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	3	1	3.0		3.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		9			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	25	1	false	true	NONE	9	9	2512.404	
	2	G235H/F170LP	NRSRAPID	10	1	true	true	NONE	9	9	1062.94	

Proposal 10961 - Observation 5 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Special Requirements	Offset 2.91 arcsec, 0.0 arcsec 5 After 3 by 0 Days to 14 Days 6 After 5 by 120 Days to 200 Days
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Proposal 10961 - Observation 4 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Observation	Proposal 10961, Observation 4: Images2										Wed May 20 18:00:46 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRCam Imaging										
	Comments: Offsets and target placement need to be adjusted based on observations acquired in JWST GO-5368 to utilize both the proposed and archival data for increased astrometric cadence. The pointing of our program should be scheduled in the Spring with 180 degree flip in PA relative to the archival data. And the positional offset should be set to provide maximal overlap with the archival data. This likely means that our observation should be offset by 1/2 a B-module field in both X and Y given that delta-PA=180. Final PAs and offsets will be entered upon completion of JWST-GO-5368 first epoch observations if possible.										
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	(Visit 4:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Template											
Dithers											
Spectral Elements											
Special Requirements											

Proposal 10961 - Observation 6 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Observation	Proposal 10961, Observation 6: Spec RV2-col1											Wed May 20 18:00:46 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: RV2 should be scheduled 5-8 months after RV1 in Cycle 4. Cycle 5 observations (e.g RV3) should be schedule 5-8 months after RV2.											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	GALCEN	RA: 17 45 40.0410 (266.4168375d)				Proper Motion RA: -2.7 mas/yr					
			Dec: -29 00 28.12 (-29.00781d)				Proper Motion Dec: -5.6 mas/yr					
			Equinox: J2000				Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Stellar associations, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	3	1	3.0		3.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		9			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	25	1	false	true	NONE	9	9	2512.404	
	2	G235H/F170LP	NRSRAPID	10	1	true	true	NONE	9	9	1062.94	

Proposal 10961 - Observation 6 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Special Requirements	Offset -2.91 arcsec, 0.0 arcsec 6 After 5 by 120 Days to 200 Days 8 After 6 by 0 Days to 14 Days
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Proposal 10961 - Observation 8 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Observation	Proposal 10961, Observation 8: Spec RV2-col2											Wed May 20 18:00:46 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	GALCEN	RA: 17 45 40.0410 (266.4168375d)				Proper Motion RA: -2.7 mas/yr					
	Dec: -29 00 28.12 (-29.00781d)				Proper Motion Dec: -5.6 mas/yr							
	Equinox: J2000				Epoch of Position: 2015.5							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Stellar associations, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	3	1	3.0		3.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	25	1	false	true	NONE	9	9	2512.404	
	2	G235H/F170LP	NRSRAPID	10	1	true	true	NONE	9	9	1062.94	

Proposal 10961 - Observation 8 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Special Requirements	8 After 6 by 0 Days to 14 Days 9 After 8 by 0 Days to 14 Days
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Proposal 10961 - Observation 9 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Observation	Proposal 10961, Observation 9: Spec RV2-col3											Wed May 20 18:00:46 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	GALCEN	RA: 17 45 40.0410 (266.4168375d) Dec: -29 00 28.12 (-29.00781d) Equinox: J2000				Proper Motion RA: -2.7 mas/yr Proper Motion Dec: -5.6 mas/yr Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	Category=Stellar Cluster											
	Description=[Stellar associations, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	3	1	3.0		3.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	25	1	false	true	NONE	9	9	2512.404	
	2	G235H/F170LP	NRSRAPID	10	1	true	true	NONE	9	9	1062.94	

Proposal 10961 - Observation 9 - Binaries In the Galactic Center - an Astrophysical Treasury Survey (BIGCATS)

Special Requirements	Offset 2.91 arcsec, 0.0 arcsec 9 After 8 by 0 Days to 14 Days
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